

DIFFERENTIAL EQUATIONS LECTURE NOTES

WEEK VII

NONHOMOGENEOUS SYSTEMS

$$\vec{x}' = A\vec{x} + \vec{p}(t) \quad \text{where } A: \text{constant matrix}$$

$$\vec{x}_p = \underbrace{\vec{x}_h}_{\substack{\text{Solution of} \\ \text{the corresponding system} \\ \vec{x}' = A\vec{x}}} + \underbrace{\vec{x}_p}_{\substack{\text{particular solution for the non-homogeneous} \\ \text{system} \\ \vec{x}' = A\vec{x} + \vec{p}(t)}}$$

To find $\vec{x}_p$ we will use the variation of parameter technique.

VARIATION OF PARAMETER (VOP) TECHNIQUE.

→ we know that $\vec{x}_h = \underline{X}(t)\vec{c}$ where $\underline{X}(t) = [\vec{x}_1(t), \dots, \vec{x}_n(t)]$ where $\vec{x}_i(t)$ are linearly independent solutions for $\vec{x}' = A\vec{x}$

→ Now, assume that $\vec{x}_p = \underline{X}(t) \cdot \vec{u}(t)$

→ unknown vector function

→ Substitute $\vec{x}_p$ in $\vec{x}' = A\vec{x} + \vec{p}(t)$. Then,

$$\vec{x}_p' = A\vec{x}_p + \vec{p}(t)$$

$$\underline{X}'(t) \cdot \vec{u}(t) + \underline{X}(t) \cdot \vec{u}'(t) = A \underline{X}(t) \vec{u}(t) + \vec{p}(t)$$

$$\underline{X}' = A\underline{X}$$

~~$$A(t)\underline{X}(t) \cdot \vec{u}(t) + \underline{X}(t) \cdot \vec{u}'(t) = A\underline{X}(t) \vec{u}(t) + \vec{p}(t)$$~~

$$\underline{X}(t) \cdot \vec{u}'(t) = \vec{p}(t)$$

$$\vec{u}'(t) = \underline{X}^{-1}(t) \vec{p}(t)$$

" $\underline{X}(t)$ is invertible
 $\det \underline{X}(t) \neq 0$ "

$$\vec{u}(t) = \int (\vec{X}^{-1}(t) \vec{p}(t)) dt$$

Then, $\vec{x}_p(t) = \vec{X}(t) \vec{u}(t) = \vec{X}(t) \int \vec{X}^{-1}(t) \vec{p}(t) dt$

∴ The general solution of Nonhomogeneous system $\vec{x}' = A\vec{x} + \vec{p}(t)$ is:

$$\vec{x}_g = \vec{x}_h + \vec{x}_p = \vec{X}(t) \vec{C} + \vec{X}(t) \int \vec{X}^{-1}(t) \vec{p}(t) dt$$

Ex Consider $\vec{x}' = \underbrace{\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}}_A \vec{x} + \underbrace{\begin{bmatrix} e^t \\ t \end{bmatrix}}_{\vec{p}(t)}$

(a) Solve $\vec{x}' = A\vec{x}$ draw phase portrait

(b) Solve $\vec{x}' = A\vec{x} + \vec{p}(t)$

(c) $\vec{x}_h$:

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & -1 \\ 3 & -2-\lambda \end{vmatrix} = \lambda^2 - 1 = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = -1 \quad \left. \begin{array}{l} \text{real distinct} \\ \text{eigenvalues.} \end{array} \right\}$$

$\lambda_1 = 1$ $(A - \lambda_1 I) \vec{v} = \vec{0}$ put $\lambda = 1 \Rightarrow (A - I) \vec{v} = \vec{0}$ $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

$$\begin{bmatrix} 1 & -1 \\ 3 & -3 \end{bmatrix} \xrightarrow{R_2 \rightarrow -3R_1 + R_2} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} v_1: \text{leading} \\ v_2: \text{free} \end{array} \quad \begin{array}{l} v_1 - v_2 = 0 \Rightarrow v_1 = v_2 \\ \vec{v} = v_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{array}$$

choose $v_2 = 1, v_1 = 0, 1$

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \vec{x}_1(t) = e^{\lambda_1 t} \vec{v} = e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda_2 = -1$ $(A + I) \vec{u} = \vec{0}$ $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$$\begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix} \xrightarrow{R_2 \rightarrow -R_1 + R_2} \begin{bmatrix} 3 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} u_1: \text{leading} \\ u_2: \text{free} \end{array}$$

$$3u_1 - u_2 = 0$$

$$u_1 = u_2/3$$

choose $u_2 = 3 \Rightarrow u_1 = 1$

(2)

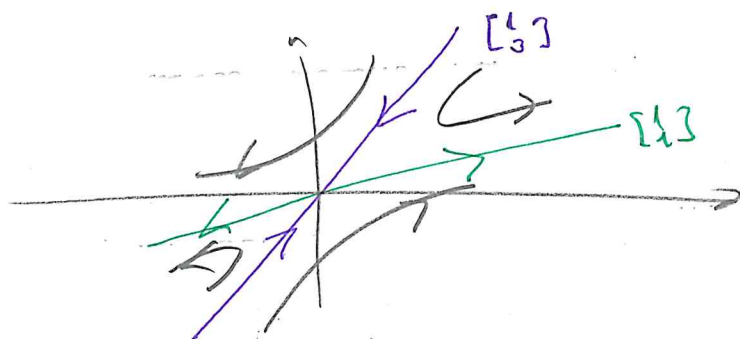
So, $\vec{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$$\vec{x}_2(t) = e^{3t} \vec{v} = e^{-t} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\vec{x}_h = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t) = c_1 e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \mathbf{X}(t) \cdot \vec{c} = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix} \vec{c}$$

$$\vec{x}_h \rightarrow \infty \text{ as } t \rightarrow \infty \quad \text{and } t \rightarrow -\infty$$

$\downarrow$
 $\vec{x}_1(t)$ dominant $\downarrow$
 $\vec{x}_2(t)$ dominant



origin is unstable dominant

⑥ Find $\vec{x}_p$ by using I.V.P

Assume $\vec{x}_p = \mathbf{X}(t) \cdot \vec{u}(t)$ $\vec{x}_p(t)$ must satisfy $\vec{x}'(t) = A \vec{x}(t) + \vec{f}(t)$

$$\cancel{\mathbf{X}'(t) \vec{u}(t)} + \mathbf{X}(t) \cdot \vec{u}'(t) = A \cancel{\mathbf{X}(t) \vec{u}(t)} + \vec{f}(t)$$

$$\boxed{\vec{u}'(t) = \mathbf{X}^{-1}(t) \vec{f}(t)}$$

$$\mathbf{X}(t) = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix} \Rightarrow \mathbf{X}^{-1}(t) = \frac{1}{\det \mathbf{X}(t)} \begin{bmatrix} 3e^{-t} & -e^{-t} \\ -e^t & e^t \end{bmatrix}$$

by part a

$$= \frac{1}{2} \begin{bmatrix} 3e^{-t} & -e^{-t} \\ -e^t & e^t \end{bmatrix}$$

$$\vec{f}(t) = \begin{bmatrix} e^t \\ t \end{bmatrix}$$

$$\vec{u}'(t) = \mathbf{X}^{-1}(t) \vec{f}(t) \Rightarrow \vec{u}(t) = \frac{1}{2} \int \begin{bmatrix} 3e^{-t} & -e^{-t} \\ -e^t & e^t \end{bmatrix} \begin{bmatrix} e^t \\ t \end{bmatrix} dt$$

$$\vec{u}(t) = \frac{1}{2} \int \begin{bmatrix} 3 - te^{-t} \\ -e^{2t} + te^t \end{bmatrix} dt = \frac{1}{2} \begin{bmatrix} \int (3 - te^{-t}) dt \\ \int (-e^{2t} + te^t) dt \end{bmatrix}$$

$$\int te^{-t} dt = -te^{-t} + \int e^{-t} dt = -te^{-t} - e^{-t} + c_1 \quad \text{choose } c_1 = 0$$

$\downarrow$
 $u = t \quad e^{-t} dt = dv$
 $dv = -e^{-t} dt \quad -e^{-t} = v$

$$\vec{v}(t) = \frac{1}{2} \begin{bmatrix} 3t + t e^{-t} + e^{-t} \\ -\frac{1}{2} e^{2t} + t \cdot e^t - e^t \end{bmatrix} \quad \therefore \vec{x}_p = \Sigma(t) \vec{v}(t) = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix} \vec{v}(t)$$

Then, $\vec{x}_p = \vec{x}_p + \vec{x}_h = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix} \vec{v}(t) + \Sigma(t) \cdot \vec{c}$
 period solution

OR we can implement Cramer-Rule (See next page) $\vec{v}(t) = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$
 $\vec{v}' = e^{-At} \vec{v}' = f(t)$

Ex 7 Solve $\vec{x}' = \underbrace{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}}_A \vec{x}(t) + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{f(t)} \quad \begin{cases} x_1' = x_1 - x_2 + 1 \\ x_2' = x_2 + 1 \end{cases} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Ans: $|A - \lambda I| = \begin{vmatrix} 1-\lambda & -1 \\ 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 = 0 \Rightarrow \lambda_{1,2} = 1$ repeated eigenvalue
 $q.p(\lambda=1) = 2$

Ans $(A - I) \vec{v} = \vec{0} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

$\begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} : \begin{matrix} v_2: \text{leading} \\ v_1: \text{free} \end{matrix} \Rightarrow \text{linearly independent vector.}$

$m(\lambda=1) = 1 < 2 = q.p(\lambda=1)$
 only one Jordan block

$-v_2 = 0 \Rightarrow v_2 = 0 \quad v_1 \in \mathbb{R} \quad \vec{v} = \begin{bmatrix} v_1 \\ 0 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ x.u.o.c. } v$
 $\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ "ordinary" eigenvector

Jordan form $J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

For generalized eigenvector $(A - \lambda I) \vec{u} = \vec{v} \quad \lambda = 1$

$(A - I) \vec{u} = \vec{v}$

$\begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -u_2 = 1 \\ u_2 = -1 \end{matrix} \left\{ \begin{matrix} u_1 = \text{free} \\ u_2 = -1 \end{matrix} \right. \vec{u} = \begin{bmatrix} u_1 \\ -1 \end{bmatrix} \text{ x.u.o.c. } u$
 $u_1 = 0$

$\vec{u} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ generalized eigenvector.

$$P = [\vec{v} \quad \vec{u}] = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow P^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow J = P^{-1}AP = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$e^{Jt} = e^t \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \Rightarrow \underset{\Phi(t)}{e^{At}} = P \cdot e^{Jt} \cdot P^{-1} = \begin{bmatrix} e^t & -t \cdot e^t \\ 0 & e^t \end{bmatrix}$$

$$\vec{x}_h(t) = e^{At} \cdot \vec{c} = \begin{bmatrix} e^t & -t \cdot e^t \\ 0 & e^t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = c_1 e^t \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -t e^t \\ e^t \end{bmatrix}$$

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ ordinary eigenvector}$$

$$A \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \vec{u} + \vec{v}$$

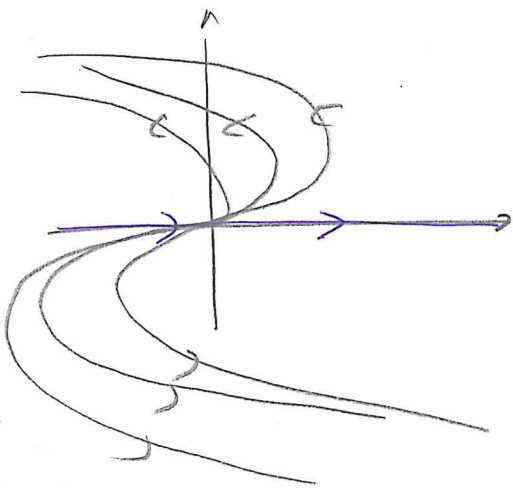
$$(A - I)\vec{u} = \vec{v} \Rightarrow A \cdot \vec{u} = \vec{u} + \vec{v} \text{ compatible with } J$$

or

$$\underline{X}(t) = e^{At} \cdot P = P \cdot e^{Jt} \cdot P^{-1} \cdot P = P \cdot e^{Jt}$$

$$\underline{X}(t) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} e^t & t e^t \\ 0 & e^t \end{bmatrix} = \begin{bmatrix} e^t & t e^t \\ 0 & -e^t \end{bmatrix}$$

$$\underline{\vec{x}}_h(t) = \underline{X}(t) \cdot \vec{c}$$



Find $\vec{x}_p(t)$

$$\underline{x}_p(t) = \underset{e^{At}}{\Phi(t)} \cdot \underset{e^{At}}{u(t)} \Rightarrow \Phi(t) \cdot u'(t) = \vec{p}(t) \text{ solve it}$$

↓
Use Cramer's

substitute in $\vec{x}' = A\vec{x} + \vec{p}(t)$

Cramer's (Cramer)

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_B \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\text{unknown}} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$x_1 = \frac{\det \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\det B}$$

$$x_2 = \frac{\det \begin{vmatrix} a_{11} & b_1 \\ a_{12} & b_2 \end{vmatrix}}{\det B}$$

Apply two technique to $\vec{X}(t) \vec{U}'(t) = \vec{p}(t)$ $\det \vec{X}(t) = e^{2t} - 0 = e^{2t}$

$$\begin{bmatrix} e^t & -te^t \\ 0 & e^t \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u_1' = \frac{\det \begin{bmatrix} 1 & -te^t \\ 1 & e^t \end{bmatrix}}{\det \vec{X}(t)} = \frac{e^t + te^t}{e^{2t}} = e^{-t} + t \cdot e^{-t}$$

$$u_1' = e^{-t} + te^{-t} \quad u_1(t) = -e^{-t} - te^{-t} - e^{-t} + d_1$$

$$u_2' = \frac{\det \begin{bmatrix} e^t & 1 \\ 0 & 1 \end{bmatrix}}{\det \vec{X}(t)} = \frac{e^t}{e^{2t}} = e^{-t} \Rightarrow u_2(t) = -e^{-t} + d_2$$

$$\vec{U}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} = \begin{bmatrix} -2e^{-t} - te^{-t} \\ -e^{-t} \end{bmatrix}$$

$$\vec{X}_p = \vec{X}(t) \cdot \vec{U} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \vec{X}_p(t) = \vec{X}_h + \vec{X}_p = \begin{bmatrix} e^t & -te^{-t} \\ 0 & e^t \end{bmatrix} \vec{c} + \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

 Solve the system

$$\vec{X}' = \begin{bmatrix} 2 & -5 \\ 1 & -2 \end{bmatrix} \vec{X} + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\vec{p}(t)}$$

$$\vec{X}_h \Rightarrow |A - \lambda I| = 0 \quad \begin{vmatrix} 2-\lambda & -5 \\ 1 & -2-\lambda \end{vmatrix} = \lambda^2 + 1 = 0 \quad \lambda_{1,2} = \pm i \text{ "Complex eigenvalues"}$$

$$\lambda_1 = i \quad (A - iI) \vec{v} = \vec{0}$$

$$\begin{bmatrix} 2-i & -5 \\ 1 & -2-i \end{bmatrix} \xrightarrow{-(2-i)R_2 + R_1} \begin{bmatrix} 0 & 0 \\ 1 & -2-i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \xrightarrow{\text{leading}} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{matrix} v_1 = (2-i)v_2 \\ v_2 = 1 \quad v_1 = 2-i \end{matrix}$$

$$\vec{v} = \begin{bmatrix} 2+i \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$e^{\lambda_1 t} \vec{v} = e^{i t} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = \underbrace{(\cos t + i \sin t)}_{e^{i t}} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$$

$$= \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} \cos t - \sin t \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) + i \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos t + \begin{bmatrix} 2 \\ 1 \end{bmatrix} \sin t \right)$$

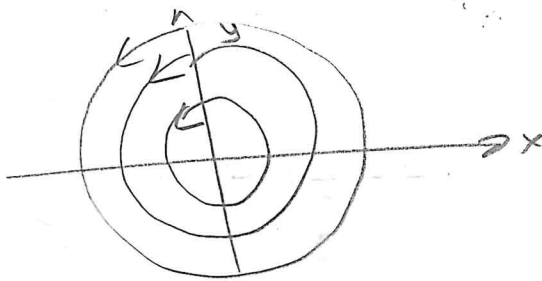
$$\vec{x}_1(t) = \operatorname{Re}(e^{\lambda_1 t} \vec{v})$$

$$\vec{x}_2(t) = \operatorname{Im}(e^{\lambda_1 t} \vec{v})$$

$$X_n = \begin{bmatrix} \vec{x}_1(t) & \vec{x}_2(t) \end{bmatrix} \vec{c} = \begin{bmatrix} 2\cos t - \sin t & \cos t + 2\sin t \\ \cos t & \sin t \end{bmatrix} \vec{c}$$

$$\underline{X}(t)$$

$$\det \underline{X}(t) = 2\cos t \sin t - \sin^2 t - \cos^2 t - 2\sin t \cos t = -1$$



$$\lambda = i = \alpha + i\beta$$

$$\alpha = \operatorname{Re}(\lambda) = 0$$

$$\beta = 1$$

$$\text{we have } \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow A \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

$$\text{Find } \vec{x}_p(t) = \underline{X}(t) \vec{u}(t) \Rightarrow \underline{X}(t) \cdot \vec{u}(t) = \vec{p}(t)$$

$$\vec{p}(t) = \begin{bmatrix} 1/t \\ 1/t \end{bmatrix}$$

Cramer's rule

$$u_1'(t) = \frac{\begin{vmatrix} 1 & \cos t + 2\sin t \\ t & \sin t \end{vmatrix}}{\det \underline{X}(t)} = \frac{-(\sin t - t \cos t - 2t \sin t)}{-1}$$

integrate

$$u_1(t) = 2\cos t + 2\sin t + t \sin t - 2t \cos t + d_1$$

$$u_2'(t) = \frac{\begin{vmatrix} 2\cos t - \sin t & 1 \\ \cos t & 1 \end{vmatrix}}{\det \underline{X}(t)} = \frac{-(2t \cos t - t \sin t - \cos t)}{-1}$$

-1

integrate

$$u_2(t) = (2-2t)\sin t - (2+t)\cos t + d_2$$

Note that if we do not take d_1 and d_2 equal to zero any constant the particular solution will give us impact the general solution, that is

$$\begin{aligned}\vec{X}_p &= \underline{X}(t) \vec{D}(t) = \underline{X}(t) \left(\begin{bmatrix} 2\cos t - 2\sin t + t\sin t - 2t\cos t \\ (2-2t)\sin t - (2+t)\cos t \end{bmatrix} + \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} \right) \\ &= \underbrace{\underline{X}(t) \vec{D}(t)}_{\vec{X}_p(t)} + \underbrace{\underline{X}(t) \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}}_{\vec{X}_h} = \vec{X}_p(t)\end{aligned}$$

Ex Find the general solution of the homogeneous system

$$\vec{X}' = \begin{bmatrix} 4 & 6 \\ -2 & -4 \end{bmatrix} \vec{X} + \begin{bmatrix} 6t \\ 0 \end{bmatrix} \quad t > 0$$

Answer

$$\vec{X}_p = \underbrace{\begin{bmatrix} 4t^3 + 3t^2 \\ -2t^3 \end{bmatrix}}_{\vec{X}_p} + \underbrace{\begin{bmatrix} 4 & 4t+1 \\ -2 & -2t \end{bmatrix} \vec{e}}_{\vec{X}_h}$$

$$\lambda = 0 \quad \text{df}(0) = 2 \quad m(\lambda = 0) = 1 \quad J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

" Jordan block